SOCIAL MEDIA PLATFORM

OBJECTIVE:

To design and implement a basic social media database system that allows users to create posts, like posts, comment on posts, follow other users, and receive notifications. The system aims to manage user interactions efficiently and enable meaningful data analysis.

DATASET DESCRIPTION:

Users Table:

Column Name	Data Type	Description
user_id	INT	Primary Key
user_name	VARCHAR	User Name
email	VARCHAR	Email Address

Posts Table:

Column Name	Data Type	Description
post_id	INT	Primary Key
user_id	INT	Foreign Key
caption	TEXT	Post Caption
posted_at	DATETIME	Post Time

Comments Table:

Column Name	Data Type	Description
comment_id	INT	Primary Key
post_id	INT	Foreign Key
user_id	INT	Foreign Key
comment_text	TEXT	Comment Content
commented_at	DATETIME	Comment Time

Likes Table:

Column Name	Data Type	Description
like_id	INT	Primary Key
post_id	INT	Foreign Key
user_id	INT	Foreign Key
liked_at	DATETIME	Like Time

Followers Table:

Column Name	Data Type	Description
follower_id	INT	Foreign Key
following_id	INT	Foreign Key
follow_date	DATETIME	Follow Date

Notifications Table:

Column Name	Data Type	Description
user_id	INT	Foreign Key
message	TEXT	Notification Message
created_at	DATETIME	Created Time

Primary Key and Foreign Key Details:

Table Name	Primary Key	Foreign Key
Users	user_id	-
Posts	post_id	user_id
Comments	comment_id	post_id, user_id
Likes	like_id	post_id, user_id
Followers	-	follower_id, following_id
Notifications	-	user_id

Entity Relationship Diagram (ERD)

Users → Posts (1:M)
Users → Comments (1:M)
Posts → Comments (1:M)
Users → Likes (1:M)
Posts → Likes (1:M)
Users → Followers (M:N self relationship)
Users → Notifications (1:M)

Descriptive Analysis:

- ➔ The total number of users on the platform was analyzed to understand the overall user base and growth of the social media application.
- ➔ The total number of posts created by users was calculated to measure content generation and platform activity levels.
- ➔ The number of likes and comments received on each post was examined to evaluate overall engagement across the platform.
- ➔ The most active users were identified based on the number of posts they created and their interaction levels.
- ➔ The average number of followers per user was computed to understand the connectivity and influence of users within the platform.
- ➔ Daily activity trends were analyzed to observe how user engagement varies over different days and time periods.

Diagnostic Analysis:

- ➔ The reasons behind certain posts receiving higher likes and comments were analyzed by comparing content type, timing, and user influence.
- ➔ Differences in user engagement levels were studied to understand why some users are more active and interactive than others.
- ➔ A decline in user activity was examined by identifying patterns and possible causes such as lack of engagement or content relevance.
- ➔ Inactive users were identified by analyzing login frequency, posting behavior, and interaction levels over time.
- ➔ The impact of different content types (text, images, videos) on user engagement was evaluated to determine what drives more interaction.
- ➔ Follower growth patterns were analyzed to understand how users gain or lose followers based on their activity and content strategy.

Predictive Analysis:

- ➔ Future user engagement was predicted based on historical activity patterns and interaction trends.
- ➔ The popularity of upcoming posts was estimated by analyzing past performance and user behavior.
- ➔ Follower growth was forecasted using existing trends in user connections and engagement levels.
- ➔ Potential influencers were identified by predicting users who are likely to gain more followers and engagement.
- ➔ Users who may become inactive in the future were predicted based on declining activity patterns.
- ➔ Future content trends were analyzed to anticipate which types of posts are likely to perform well.

Prescriptive Analysis:

- ➔ The best time for users to post content was suggested based on peak activity and engagement trends.
- ➔ Content strategies were recommended to users to improve visibility and increase engagement on their posts.
- ➔ Methods to enhance user engagement, such as interactive content and consistent posting, were proposed.
- ➔ Strategies to increase follower count were suggested, including regular posting and engaging with followers.
- ➔ A content recommendation approach was designed to suggest relevant posts to users based on their interests.
- ➔ Notification strategies were optimized to ensure users receive timely updates that encourage interaction and platform usage.

CONCLUSION

This project successfully demonstrates how SQL can be used to analyze and manage data in a social media platform. By working with multiple tables such as users, posts, comments, likes, followers, and notifications, meaningful insights were derived about user behavior and platform activity. The analysis helped in identifying active users, popular posts, and engagement patterns across the platform. It also highlighted the importance of relationships between tables in understanding real-world scenarios.

Through descriptive, diagnostic, predictive, and prescriptive analysis, the project provided valuable recommendations to improve user engagement and platform performance. Overall, this project shows how structured data and SQL queries can be effectively used to support data-driven decisions and enhance the functionality of a social media system.